Cloud Services in AWS



Services, Use Cases, and Architecture

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Agenda

1. **AWS Core Infrastructure**
2. **AWS AI Services**
3. **AWS Security & Compliance**
4. **AWS Resilience & DevOps**
5. **Architecture & Conclusions**

Session Goal

Understand how AI is used in the cloud and recognize the main **AWS** services: what they do and where they fit in a real-world architecture.

120 minutes | Engineering





AWS Core Infrastructure

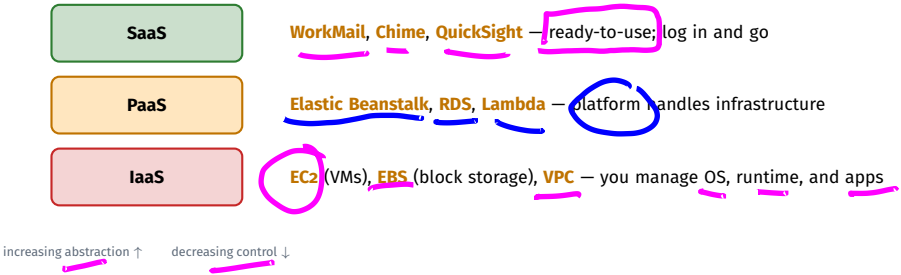
Part 1





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AWS Service Models: IaaS, PaaS, SaaS



As abstraction increases you **manage less** but also **control less**. Choose based on team expertise, compliance needs, and workload type.



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AWS Pricing Basics & Free Tier

Free Tier (accounts since July 2025)

- **Free plan**: $100 in credits + up to $100 more by completing activities; protected from charges for 6 months or until credits run out
- **Paid plan**: same credits, then pay as you go
- **Always free**: 30+ services with monthly limits (e.g. Lambda 1M requests, DynamoDB 25 GB)

Three Ways AWS Charges

- **Pay as you go**: per second, GB, request, or token
- **Save by committing**: Savings Plans and Reserved capacity (1–3 years)
- **Pay less as you use more**: tiered volume discounts

Accounts opened before July 15, 2025 keep the older 12-month Free Tier.

⚠ Create an **AWS Budget** alert on day one: a forgotten NAT Gateway, endpoint, or EC2 instance is the most common surprise bill.



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AWS Core Services Map

Building blocks: combine them to run any workload

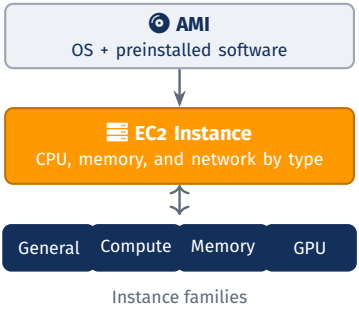
COMPUTE	STORAGE	DATABASES	NETWORKING
EC2 · Lambda	**S3 · EBS · EFS**	**RDS · DynamoDB**	**VPC · Route 53**

Foundations: where everything runs and how it is protected

GLOBAL INFRASTRUCTURE	CROSS-CUTTING
Regions · Availability Zones · Edge	**Security · Monitoring · DevOps**

Key idea: AI services are built on top of these blocks; understand them first.

Amazon EC2: Virtual Servers in the Cloud



Use Cases

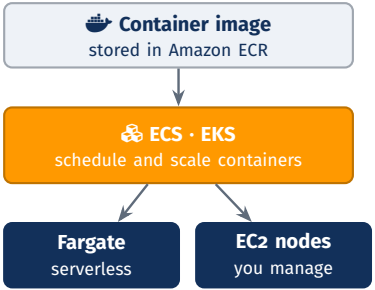
- Web and application servers
- Lift-and-shift migrations
- Training and inference on GPU instances
- Batch and high-performance computing

On-Demand | **Reserved** | **Savings Plans** | **Spot**

Spot is up to 90 % cheaper, but AWS can reclaim the instance with a 2-minute notice.

EC2 gives full control of the server: you choose the OS, size, and pricing model — and you operate it.

Containers on AWS: ECS, EKS & Fargate



Use Cases

- Microservices and APIs
- Portable apps: same image everywhere
- Kubernetes-based ML and batch workloads
- CI/CD friendly deployments

ECS: AWS-native **EKS: Kubernetes**

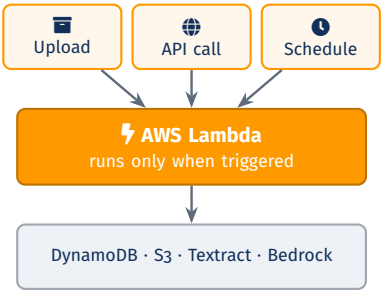
Fargate: no servers to manage

A container packages the app with its dependencies; **Fargate** removes the need to manage the machines that run it.



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AWS Lambda: Serverless Functions



Use Cases

- Glue code between AWS services
- File processing when an object is uploaded
- Serverless APIs behind API Gateway
- Orchestrating AI pipelines

Maximum runtime is **15 minutes**: not for long-running or stateful workloads.

Lambda scales automatically **to zero**: you pay only while your code runs.

Choosing Compute: EC2 vs. Containers vs. Lambda

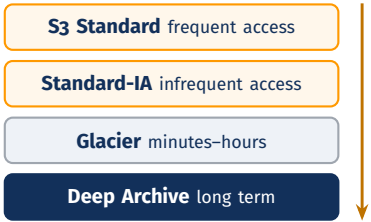
	EC2	ECS / EKS	Lambda
Unit	Virtual machine	Container	Function
You manage	OS, runtime, scaling	Images, task definitions	Code only
Scaling	Auto Scaling groups	Service autoscaling	Automatic, to zero
Billing	Per second while running	Per task or node	Per invocation
Typical use	Legacy apps, GPUs	Microservices	Event-driven glue

EC2: maximum control. **Containers**: portable workloads. **Lambda**: zero management for short, event-driven tasks.



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Amazon S3: Object Storage for Everything



Lifecycle rules move objects to cheaper classes

Use Cases

- Data lakes and ML datasets
- Backups and compliance archives
- Static websites and media
- Input and output of AI services

11 nines durability **Versioning**

Encryption **Event notifications**

S3 is the default landing zone for data: most AWS AI services read from and write to **S3 buckets**.



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Choosing Storage: S3 vs. EBS vs. EFS

	S3	EBS	EFS
Type	Object	Block	Shared file system (NFS)
Access	Any app through the API	One EC2 instance	Many instances at once
Performance	Web-scale throughput	Highest IOPS (SSD)	Elastic, multi-AZ
Typical use	Data lakes, backups	Databases, boot volumes	Shared content, ML datasets

Which storage type would you choose for a high-throughput database vs. a long-term document archive? Why?





AWS Databases: Choosing the Right Engine

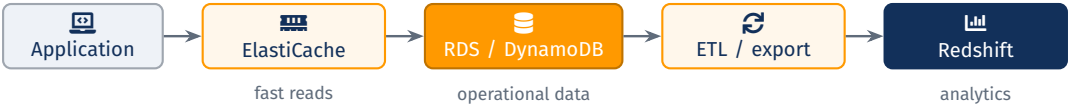
	RDS / Aurora	DynamoDB	ElastiCache	Redshift
Model	Relational (SQL)	NoSQL key-value, document	In-memory (Redis)	Columnar warehouse
Strength	ACID, Multi-AZ failover	Single-digit ms at any scale	Sub-millisecond reads	SQL over petabytes
Management	Managed engine	Serverless	Managed cluster	Managed cluster
Typical use	E-commerce, ERP	Gaming, IoT, sessions	Cache, leaderboards	BI and reporting

There is no single best database: pick the engine by **access pattern**, **consistency**, and **scale**.





Databases in an Application Architecture



Operational (OLTP)

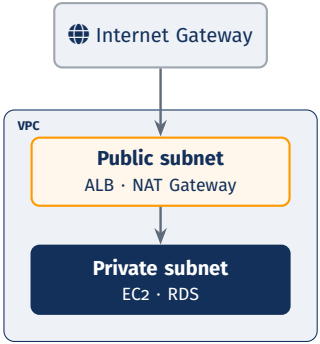
- Many small reads and writes
- Low latency, high availability
- Cache in front of the database

Analytical (OLAP)

- Large scans and aggregations
- Historical data, BI dashboards
- Also feeds ML and AI pipelines

Separate **operational** from **analytical** workloads so reports never slow down the application.

Amazon VPC: Your Private Network



Key Concepts

- Choose your own IP range (CIDR)
- Public subnets: load balancers, NAT
- Private subnets: apps and databases
- Security Groups and NACLs filter traffic
- VPC endpoints reach S3 or Bedrock without the internet

Never expose databases directly to the internet: keep them in private subnets.

Reaching Users: Route 53, CloudFront & ELB



Route 53

DNS and health checks. Weighted, latency, and failover routing.

CloudFront

Global CDN with 400+ points of presence. Caches content near users.

ELB / ALB / NLB

Distributes traffic across healthy targets at layer 7 or layer 4.

Route 53 resolves names; **CloudFront** serves content from the nearest edge; **ELB** spreads the load.



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Pricing: Compute

Service	Billed by	Indicative price	Free tier
EC2 On-Demand	Per second	t3.micro ≈ $0.0104/h (about $7.6/month)	Credits (Free plan eligible types)
Spot / Savings Plans	Interruptible or 1–3 yr commitment	Spot up to 90 % off; plans up to ≈72 % off	None
Fargate	vCPU-h and GB-h	≈ $0.040/vCPU-h + $0.0044/GB-h	Credits
EKS	Cluster-hour + nodes	$0.10/cluster-hour (about $73/month) + compute	None
Lambda	Requests + GB-seconds	$0.20 per 1M requests + $0.0000167 per GB-s	**Always free**: 1M requests + 400 000 GB-s/month

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Pay for what runs: **Lambda** and **Spot** cost least for bursty work; **idle EC2 and EKS** keep billing.





Cost Comparison: One Month of Compute

Option	Configuration	Calculation	Approx. monthly cost
EC2	t3.micro, running 24/7	730 h × $0.0104	**$7.6**
Fargate	0.25 vCPU + 0.5 GB, 24/7	730 h × $0.0123	**$9**
EKS	Control plane only (no nodes)	730 h × $0.10	**$73**
Lambda	1M requests, 128 MB, 200 ms	$0.20 + 25 000 GB-s × $0.0000167	**$0.62** (free tier: $0)
Lambda	10M requests, same function	$2.00 + 250 000 GB-s × $0.0000167	**$6.2**

Calculation assumes 730 hours per month. Indicative US East list prices; check aws.amazon.com/pricing.

Lambda wins for **sporadic** traffic; a small always-on EC2 or Fargate task is cheaper once load is **constant**.



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Pricing: Storage

Service	Billed by	Indicative price	Free tier
S3 Standard	GB-month + requests	≈ $0.023/GB-month (first 50 TB)	Credits; legacy accounts: 5 GB for 12 months
S3 Standard-IA	GB-month + retrieval	≈ $0.0125/GB-month	None
Glacier Deep Archive	GB-month + retrieval	≈ $0.001/GB-month; hours to restore	None
EBS gp3	Provisioned GB-month	≈ $0.08/GB-month	Legacy: 30 GB for 12 months
EFS Standard	GB-month used	≈ $0.30/GB-month	Legacy: 5 GB for 12 months

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Match data to the **right class**: lifecycle rules can cut archive cost by more than 95 %.





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Cost Comparison: Storing 1 TB for a Month

Service	Best for	Access time	Approx. cost / TB-month
S3 Standard	Active data, data lakes	Milliseconds	**$23**
S3 Standard-IA	Backups read rarely	Milliseconds (+ retrieval fee)	**$12.5**
Glacier Deep Archive	Compliance archives	Hours	**$1**
EBS gp3	Databases, boot volumes	Lowest latency (one EC2)	**$80**
EFS Standard	Shared file system	Low latency (many EC2)	**$300**

Storage only; requests and retrieval are billed separately. Indicative US East list prices; check aws.amazon.com/pricing.

The same terabyte costs from **$1 to $300**: the access pattern decides the service, and the price follows.





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Pricing: Databases

Service	Billed by	Indicative price	Free tier
RDS (db.t3.micro)	Instance-hour + storage	≈ $0.017/h + $0.115/GB-month; Multi-AZ about 2×	Credits; legacy: 750 h for 12 months
DynamoDB on-demand	Requests + storage	≈ $0.625/M writes, $0.125/M reads, $0.25/GB-month	**Always free**: 25 GB storage
ElastiCache (t3.micro)	Node-hour	≈ $0.017/h per node	Credits
Redshift Serverless	RPU-hour used	≈ $0.375 per RPU-hour	Trial credits

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Cost follows the **access pattern**: pay per request (DynamoDB) or per hour provisioned (RDS, ElastiCache).





Cost Comparison: Small Production Database

Option	Configuration	Calculation	Approx. monthly cost
RDS Single-AZ	db.t3.micro + 20 GB	730 h × $0.017 + 20 GB × $0.115	**$15**
RDS Multi-AZ	Same, with standby	About 2× instance and storage	**$29**
DynamoDB	1M writes + 5M reads, 1 GB	$0.63 + $0.63 (storage in free 25 GB)	**$1.3**
ElastiCache	cache.t3.micro, 1 node	730 h × $0.017	**$12**
Redshift Serverless	8 RPU, 2 h/day	60 h × 8 × $0.375	**$180**

Calculation assumes 730 hours per month. Indicative US East list prices; check aws.amazon.com/pricing.

DynamoDB is cheapest at low volume; **provisioned** engines pay for idle hours but become predictable at scale.





Pricing: Networking

Service	Billed by	Indicative price	Free tier
VPC, subnets, SGs	No charge	The network itself is free	Always free
NAT Gateway	Hour + GB processed	≈ $0.045/h + $0.045/GB	None
ALB	Hour + LCU	≈ $0.0225/h + $0.008 per LCU-hour	Credits
Route 53	Zone + queries	$0.50/hosted zone/month + $0.40/M queries	None
CloudFront	GB out + requests	≈ $0.085/GB after the free allowance	**Always free**: 1 TB + 10M requests/month
Data transfer out	GB to the internet	≈ $0.09/GB; inbound is free	**Always free**: 100 GB/month

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Hidden cost: **NAT Gateways** and **data transfer** often surprise first bills.



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Cost Comparison: Network Options

Option	Fixed cost (24/7)	Variable cost	Example scenario
NAT Gateway	$33/month	$0.045/GB	500 GB: ≈ **$55**
S3 Gateway endpoint	Free	Free	500 GB to S3: **$0**
Interface endpoint	≈ $7/AZ	$0.01/GB	500 GB, 1 AZ: ≈ **$12**
ALB	≈ $16/month	$0.008 per LCU-hour	Light traffic: ≈ **$20**
Internet egress	Free (100 GB)	≈ $0.09/GB	1 TB out: ≈ **$81**
CloudFront	Free (1 TB)	≈ $0.085/GB after	1 TB out: **$0**

Fixed cost assumes 730 hours per month. Indicative US East list prices; check aws.amazon.com/pricing.

Use **VPC endpoints** instead of NAT for S3 traffic, and **CloudFront** instead of direct egress.





AWS AI Services

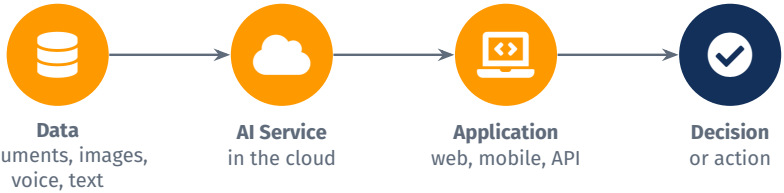
Part 2





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AI in the Cloud: Why Does It Matter?



Cloud AI is **not just a chatbot**: it includes services for **documents**, **images**, **voice**, **language**, **machine learning**, and **generative AI**.

Why Use AI in the Cloud?

Advantages

- Scales on demand
- Pay-per-use model
- Pre-trained models and services ready
- Integrates with apps and data pipelines
- High availability built-in
- Less infrastructure to manage
- Add AI capabilities **quickly**

Challenges

- Costs grow with usage
- Vendor dependency risk
- Data privacy and governance
- Latency and connectivity
- Model biases and errors
- Regulatory compliance (GDPR, HIPAA)

The cloud **facilitates** AI access, but **does not eliminate** cost, security, governance, and quality challenges.



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AWS AI Services Map

Specialized services: solve a specific, well-defined problem

DOCUMENTS
Textract

VISION
Rekognition

VOICE
Transcribe · Polly

LANGUAGE
Comprehend

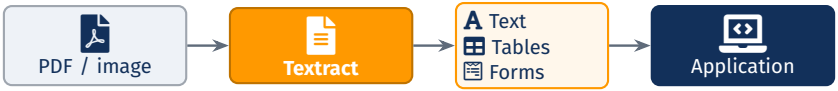
Platforms: for building broader, custom solutions

GENERATIVE AI
Amazon Bedrock

MACHINE LEARNING
Amazon SageMaker

Key idea: first identify the **problem**; then choose the service.

Amazon Textract: Documents Become Data



Use Cases

- **Invoices**
- **Forms**
- **Contracts**
- **Banking documents**
- **Loan applications**
- **Government documents**

*What would you do with the information **after** extracting it with Textract?*

Textract adds **document intelligence** to an application without building OCR and structure recognition from scratch.

Amazon Rekognition: AI for Images & Video

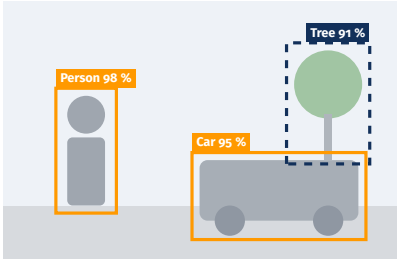


Image → Rekognition → labels + location + confidence

Use Cases

- Object and scene detection
- Image and video analysis
- Content moderation at scale
- Facial analysis **only where legal and ethical**

⚠ Facial recognition requires consent, legal basis, and careful handling of potential biases.

Computer vision is consumed as a cloud API — no GPU cluster required.

Amazon Transcribe + Amazon Polly

Transcribe
Voice → Text

Polly
Text → Voice

They are **opposites**: one listens, the other speaks.

Together they form a complete voice assistant pipeline:



Voice assistants | **Call centers** | **Meeting transcripts** | **Accessibility** | **Customer service**

Multiple AI capabilities **combine** to build a complete end-to-end experience.

Amazon Comprehend: Meaning from Text



What it can detect

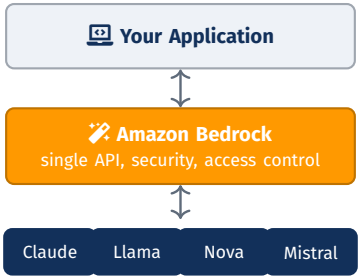
- Sentiment: positive, negative, neutral, mixed
- Entities: people, places, dates, organizations
- Text classification into custom categories
- Topic modeling
- Personal information (PII) detection

Use Cases

Customer feedback **Support tickets** **Document classification**

Language processing converts **raw text** into **structured information** that an application can act on automatically.

Amazon Bedrock: The Generative AI Platform



Foundation models from multiple providers

Use Cases

- Enterprise chatbots and assistants
- Summarization and content generation
- Q&A over your own documents (RAG)
- Agents that plan and execute tasks

Bedrock is **not a model**: it is a platform for building applications using multiple foundation models.

Bedrock: Key Concepts

Foundation Model
Large, pre-trained, general-purpose model.

LLM
Foundation model specialized in language.

Prompt
The instruction we give to the model.

Embeddings
Text as numbers capturing semantic meaning.

RAG
Retrieve your own docs and give them to the model.

Knowledge Bases
Managed RAG service inside Bedrock.

Agents
Model plans steps and uses tools.

Guardrails
Safety filters and business policy rules.

Bedrock lets you embed generative AI in enterprise applications **without building a foundation model from scratch**.





SageMaker vs. Bedrock

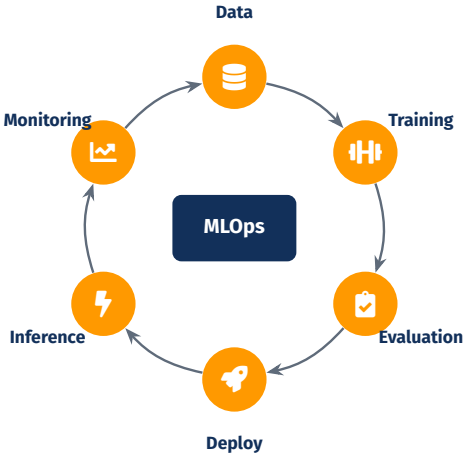
	Bedrock	SageMaker
Focus	Generative AI	General machine learning
Models	Ready-to-use foundation models	Custom or pre-existing models
Training	Not the main focus	**Yes**, its core strength
Customization	RAG, fine-tuning, Guardrails	Full control, any framework
Typical use	Build generative apps fast	Develop & operate custom ML

Bedrock: build generative applications quickly with existing models. **SageMaker**: develop, train, and operate your own ML models end-to-end.





SageMaker: The ML Model Lifecycle



- **Prepare data**: clean and organize.
- **Train**: model learns from labeled data.
- **Evaluate**: measure accuracy and bias.
- **Deploy**: publish as a REST endpoint.
- **Inference**: use it to predict in production.
- **MLOps**: automate and monitor the full cycle.





Pricing: Specialized AI Services

Service	Billed by	Indicative price	Free tier
Textract	Per page	Text ≈ $1.50 per 1 000 pages; tables and forms cost more	3 months: 1 000 pages/month (Paid plan)
Rekognition	Per image / video minute	≈ $1.00 per 1 000 images (first 1M)	Check AWS page; older accounts: 5 000 images/month
Transcribe	Per audio second	≈ $0.024 per minute	12 months: 60 min/month (Paid plan)
Polly	Per character	Standard $4, Neural $16 per 1M characters	Credits
Comprehend	Per 100-character unit	≈ $0.0001 per unit	12 months: 50K units per API/month (Paid plan)

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

APIs are cheap per call but **volume** adds up: estimate pages, images, or minutes per month first.





Cost Comparison: AI Services for a Real Workload

Service	Workload	Unit price	Approx. cost
Textract	10 000 scanned pages (text)	$1.50 per 1 000 pages	**$15**
Rekognition	10 000 images analyzed	$1.00 per 1 000 images	**$10**
Transcribe	100 hours of audio	$0.024 per minute	**$144**
Polly Neural	1M characters (about 100 pages)	$16 per 1M characters	**$16**
Comprehend	1M characters of text, one API	$0.0001 per 100-char unit	**$1**
Bedrock	1M tokens	Depends on the model	See the model price list

Example workloads are illustrative. Indicative US East list prices; check aws.amazon.com/pricing.

Audio is the **most expensive** input per hour; text analysis is almost free. Estimate volume before choosing.





Pricing: Bedrock & SageMaker

Service	Billed by	Indicative price	Free tier
Bedrock on-demand	Input and output tokens	Per 1 000 tokens; varies by model and provider	Credits
Bedrock provisioned	Model units per hour	Hourly commitment for guaranteed throughput	None
Batch inference	Tokens, discounted	Lower rate for non-urgent jobs (about 50 % less)	None
SageMaker notebooks	Instance-hour	ml.t3.medium ≈ $0.05/h while running	Legacy: 250 h for 2 months
Training / endpoints	Instance-hour (or per ms)	Depends on instance type, e.g. GPU costs dollars per hour	None

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Estimate **tokens** for Bedrock; stop **idle endpoints** in SageMaker — they bill until deleted.





AWS Security & Compliance

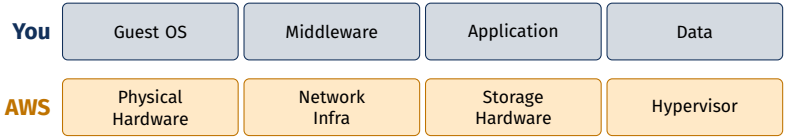
Part 3





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Shared Responsibility Model



AWS: security *of* the cloud

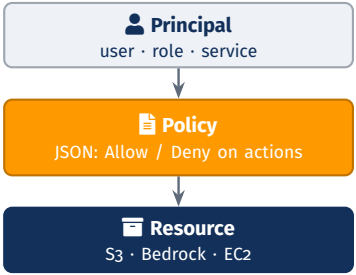
Data centers, hardware, network, and the virtualization layer. Compliance certifications (ISO 27001, SOC 2).

You: security *in* the cloud

Identities, permissions, data, encryption, network rules, and application code.

AWS secures the platform; **you** configure it securely. Most cloud incidents come from **misconfiguration**, not from AWS failures.

AWS IAM: Who Can Do What



STS issues temporary credentials

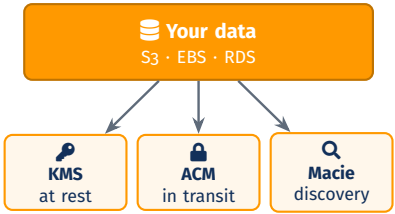
Best Practices

- Least privilege: minimum permissions
- Roles, not long-lived access keys
- MFA on every privileged account
- Audit with IAM Access Analyzer

Never put access keys in code or repositories.

In AWS every request is evaluated against **policies**: identity is the **new perimeter**.

Protecting Data: KMS, ACM & Macie



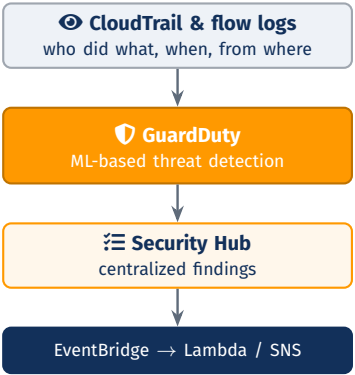
Key Ideas

- Encrypt at rest and in transit by default
- Managed keys with rotation and full audit
- Free TLS certificates with automatic renewal
- Detect PII in S3 before it reaches an AI model

Secrets Manager **Comprehend PII**

Know **where sensitive data lives** and keep it **encrypted** before sending it to any AI service.

Detecting Threats: CloudTrail, GuardDuty & Security Hub



What Each Service Does

- **CloudTrail**: records every API call
- **GuardDuty**: flags unusual behavior
- **Security Hub**: one view of all findings
- **EventBridge**: triggers automatic response

> *If an access key leaks at 3 am, which service tells you first?*

Defense in Depth: Security Layers on AWS

Layer	Services	Protects against
Edge	Shield, CloudFront	Volumetric DDoS attacks
Application	AWS WAF	SQL injection, XSS, bots, rate abuse
Network	Security Groups, NACLs	Unwanted ports and CIDR ranges
Identity	IAM, STS	Excessive or stolen permissions
Data	KMS, Macie, ACM	Exposure of sensitive data
Detection	CloudTrail, GuardDuty	Unnoticed attacks and misconfigurations

No single control is enough: **layer** identity, network, data, and detection so one failure does not become a breach.



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Pricing: Security Services

Service	Billed by	Indicative price	Free tier
IAM	No charge	Free for every account	**Always free**
KMS	Key-month + requests	$1 per key/month + $0.03 per 10 000 requests	**Always free**: 20 000 requests/month
ACM	Public certificates	Free for public certificates	**Always free**
CloudTrail	Events delivered	Event history (90 days) free; extra trails $2 per 100 000 events	First management-events trail free
GuardDuty / Macie	Data analyzed	Volume-based after the trial	30-day free trial
AWS WAF	ACL + rules + requests	$5/web ACL + $1/rule per month + $0.60 per 1M requests	None

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Core identity and encryption features are **free or cheap**: security is rarely a cost excuse.





AWS Resilience & DevOps

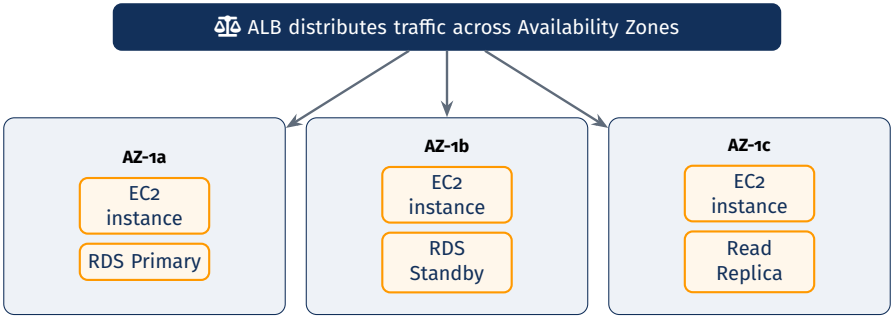
Part 4





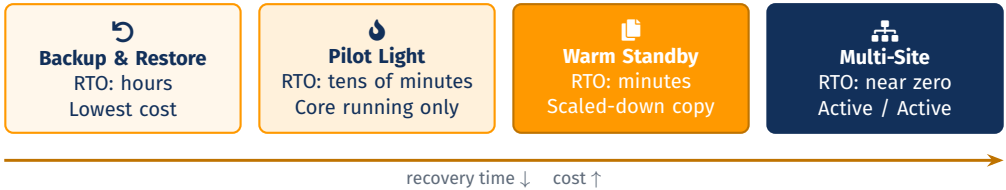
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High Availability on AWS: Multi-AZ & Multi-Region



Multi-AZ gives **automatic failover** in seconds. For extreme resilience, combine with **multi-region** replication and Route 53 health-check routing.

Disaster Recovery Strategies



RTO: Recovery Time Objective

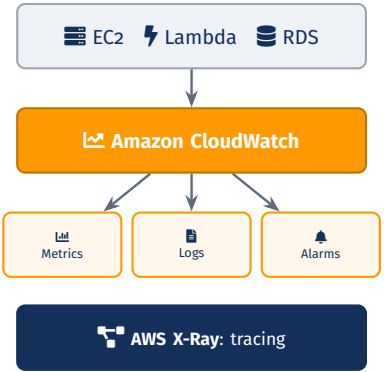
How long the business can be down.

RPO: Recovery Point Objective

How much data loss is acceptable.

Define **RPO and RTO** before disaster strikes, choose the cheapest strategy that meets them, and **test recovery** regularly.

AWS Monitoring: CloudWatch & X-Ray



Three Pillars of Observability

- **Metrics**: CPU, requests/sec, latency
- **Logs**: application events and errors
- **Traces**: request flow across services

SLI / SLO / SLA

- **SLI**: measured uptime (CloudWatch)
- **SLO**: your internal target (e.g. 99.9 %)
- **SLA**: AWS contractual commitment

AWS CI/CD: From Commit to Production



AWS CodePipeline orchestrates all stages end to end

Why Automate

- Every change is built and tested
- Security scans before deploy
- Fast, repeatable releases

Safe Deployment Patterns

Blue/Green **Canary** **Rolling**

Roll back automatically if health checks fail.

A pipeline turns deployment into a **repeatable, auditable** process instead of a manual risk.

Infrastructure as Code: CloudFormation, CDK & Terraform

	CloudFormation	AWS CDK	Terraform
Language	YAML / JSON	Python, TypeScript, Java	HCL
Scope	AWS only	AWS (synthesizes CFN)	Multi-cloud
State	Managed by AWS	Managed by AWS	State file (S3 backend)
Best for	Declarative AWS stacks	Developers; reusable code	Hybrid or multi-cloud

Infrastructure defined as code is **reproducible**, **version-controlled**, and **auditable** — the same stack can be recreated in any account or Region.





Pricing: Resilience & DevOps

Service	Billed by	Indicative price	Free tier
CloudWatch	Metrics, logs, alarms	Logs $0.50/GB ingested; custom metrics $0.30/month; alarms $0.10/month	**Always free**: 10 metrics, 10 alarms, 5 GB logs
X-Ray	Traces recorded	$5 per 1M traces recorded	**Always free**: 100 000 traces/month
CodePipeline	Pipeline / action-minute	V1 $1 per active pipeline/month; V2 per action-minute	1 active pipeline free
CodeBuild	Build minute	$\approx$ $0.005/min (small Linux)	**Always free**: 100 build min/month
CloudFormation / CDK	Resources created	No extra charge; you pay for the resources	Always free
Multi-AZ and DR	Duplicated resources	Multi-AZ RDS about 2$\times$; Elastic Disaster Recovery $\approx$ $0.028/server-hour	None

Indicative US East (N. Virginia) list prices; they change often. Check aws.amazon.com/pricing.

Resilience is **paid in duplicated resources**: pick the DR strategy your RPO and RTO justify.

Architecture & Conclusions

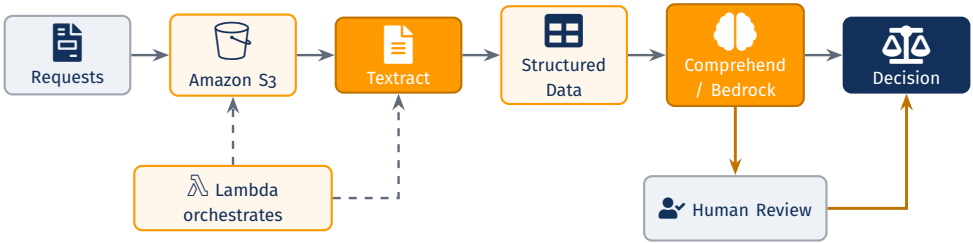
Part 5





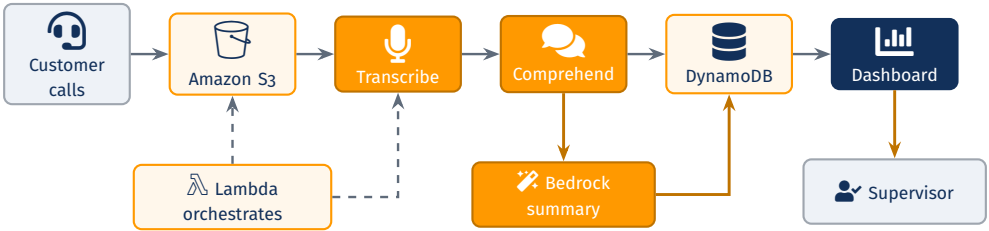
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Case Study 1: Intelligent Document Processing



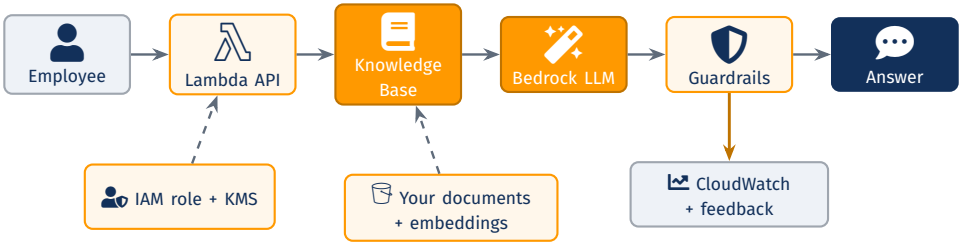
Real value appears when services are **integrated into an architecture**, not when used in isolation.

Case Study 2: Call-Center Analytics



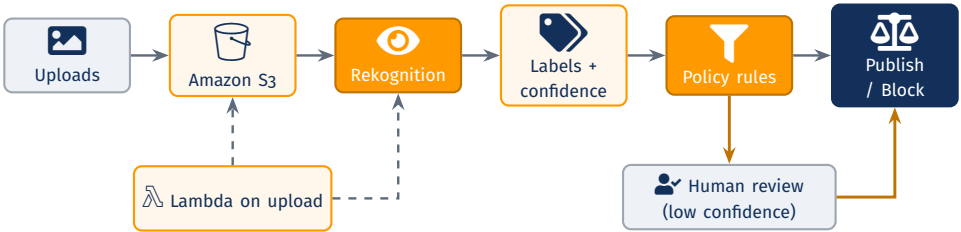
Voice becomes text, text becomes **sentiment and PII flags**, and a generative summary lets supervisors act on **negative calls** first.

Case Study 3: Enterprise Chatbot with RAG



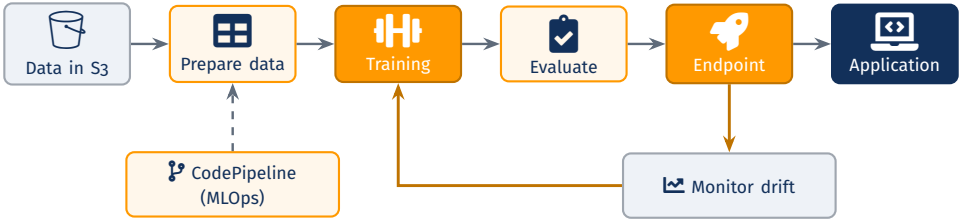
The model answers from **your own documents**, with **guardrails** on the output — no foundation model is trained or fine-tuned.

Case Study 4: Image Moderation Pipeline



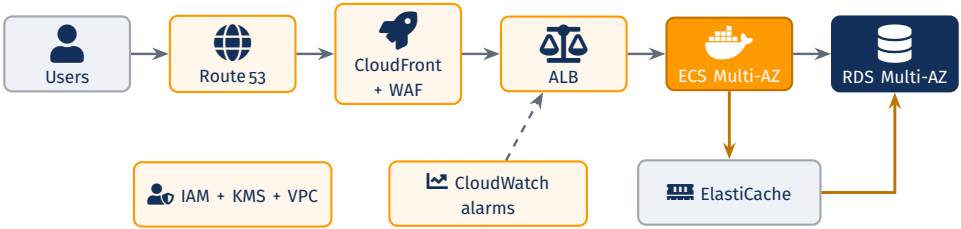
Automatic moderation handles the **clear cases**; only **uncertain images** reach a person, which controls cost and bias.

Case Study 5: Custom ML Model with SageMaker



Training is only **one stage**: monitoring drift and **retraining automatically** is what keeps a model useful in production.

Case Study 6: Resilient Web Application



Edge protection, **multi-AZ compute and database**, a cache, and **alarms** combine to survive failures without manual action.

Think Together (1/3): Infrastructure

Question 1

A job runs for 3 minutes each time a file is uploaded. Would you use **EC2, containers, or Lambda**? Why?

Question 2

Which **database** fits session data with millisecond latency, and which fits monthly reports?

Question 3

Where would you keep **5 TB of rarely read** compliance PDFs, and where a database volume?

Question 4

Why should the database live in a **private subnet** and S3 traffic use a VPC endpoint?

Hint: compare control, operations effort, access pattern, and cost for each option.





Think Together (2/3): AI Services

Question 5

Which AWS service would you use to **extract** data from scanned documents?

Question 6

At what point would it make sense to add a **generative LLM** via Bedrock?

Question 7

Which would you use to **analyze** sentiment and entities in extracted text?

Question 8

Where should **human validation** exist in the pipeline? Why?

Hint: there is no single correct answer; it depends on the risk level of each decision.





Think Together (3/3): Security, Resilience & Cost

Question 9

An app must read from S3: **user access keys or an IAM role**? What are the risks of each?

Question 10

Your RPO is 15 minutes and RTO 1 hour. Is **Multi-AZ** enough, or do you need another DR strategy?

Question 11

Name the **security layers** that protect a public API, from the edge to the data.

Question 12

Which **three numbers** would you estimate before launch to avoid a surprise bill?

Hint: think about failure, attack, and cost before the first deployment, not after.



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Ideas to Remember (1/2): Foundations

1. Core infrastructure is a set of **building blocks**: EC2 gives control, containers give portability, Lambda removes management; **S3** is the default home for data.

2. Match storage and databases to the **access pattern**: object, block, or file; SQL, NoSQL, cache, or warehouse. The same terabyte can cost **$1 or $300**.

3. Cloud cost is **pay as you go**: use free tiers, estimate volume first, set budget alerts, and watch **NAT, data transfer, and idle resources**.





Ideas to Remember (2/2): AI, Security & Operations

1. AWS offers **specialized AI services** and **platforms** (Bedrock, SageMaker); the right one depends on the **problem**, data, cost, and risk.

2. Security is **shared**: apply least-privilege **IAM**, encrypt data, detect threats, and layer defenses from the edge to the data.

3. Resilience and delivery are designed in: **Multi-AZ**, DR by RPO and RTO, **observability**, CI/CD, and IaC. Value comes from **integrated architectures**.

Which AWS service would you use to solve a real problem in your own field of study or work?



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Thanks!

Questions

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